

✧ Exercise No. 8.4 ✧

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Q1 Prove that the planes $4x + 4y - 5z = 12$,
 $8x + 12y - 13z = 32$ intersect & eqs. of their line
of intersection can be written in the form

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z}{4}$$

Soln. Given planes are

$$4x + 4y - 5z = 12 \quad \text{--- ①}$$

$$8x + 12y - 13z = 32 \quad \text{--- ②}$$

d.s. of the normal to plane ① are 4, 4, -5

d.s. of the normal to plane ② are 8, 12, -13

Since d.s. of both normals are not proportional

Hence both normals are not parallel

So given planes are not parallel

Hence given planes intersect.

Now we find eq. of their
line of intersection.

For this Put $z = 0$ in ① & ②

$$4x + 4y - 12 = 0$$

$$8x + 12y - 32 = 0$$

$$\frac{x}{-12+14} = \frac{y}{-12+16} = \frac{1}{48-32}$$

$$\frac{x}{16} = \frac{y}{32} = \frac{z}{16}$$

$$\boxed{x=1, y=2}$$

So, $(1, 2, 0)$ is a pt. on line of intersection

Again Put $x=0$ in ① & ②

$$\left. \begin{aligned} 4y - 5z - 12 &= 0 \\ 12y - 13z - 32 &= 0 \end{aligned} \right\}$$

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$$\frac{y}{160-156} = \frac{-z}{-128+144} = \frac{1}{-52+60}$$

$$\frac{y}{4} = \frac{z}{-16} = \frac{1}{8}$$

$$\Rightarrow \boxed{y = \frac{1}{2}, z = -2}$$

So, $(0, \frac{1}{2}, -2)$ is another pt. on line of intersection

Hence the req. eq. of line through $(1, 2, 0)$ & $(0, \frac{1}{2}, -2)$

$$\text{is } \frac{x-1}{0-1} = \frac{y-2}{\frac{1}{2}-2} = \frac{z-0}{-2-0}$$

$$\frac{x-1}{-1} = \frac{y-2}{-\frac{3}{2}} = \frac{z}{-2}$$

$$\frac{x-1}{-2} = \frac{y-2}{-3} = \frac{z}{-4}$$

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z}{4} \text{ is req. eq.}$$

Q2 Find symmetric form for the line

$$x+y+z+1=0 = 4x+y-2z+2$$

Sol. Given eq. of line is

$$x+y+z+1=0 \quad \text{--- ①}$$

$$4x+y-2z+2=0 \quad \text{--- ②}$$

Put $z=0$ in ① & ②

$$\left. \begin{array}{l} x+y+1=0 \\ 4x+y+2=0 \end{array} \right\}$$

$$\frac{x}{2-1} = \frac{-y}{2-4} = \frac{1}{1-4}$$

$$\frac{x}{1} = \frac{y}{2} = \frac{1}{-3}$$

$$\Rightarrow x = -\frac{1}{3}, y = -\frac{2}{3}$$

∴ A $(-\frac{1}{3}, -\frac{2}{3}, 0)$ is a pt. on the given line.

Now Put $x=0$ in ① & ②

$$\left. \begin{array}{l} y+z+1=0 \\ y-2z+2=0 \end{array} \right\}$$

$$\frac{y}{2+2} = \frac{-z}{2-1} = \frac{1}{-2-1}$$

$$\frac{y}{4} = \frac{z}{-1} = \frac{1}{-3}$$

$$\Rightarrow y = -\frac{4}{3}, z = \frac{1}{3}$$

∴ B $(0, -\frac{4}{3}, \frac{1}{3})$ is another pt. on given line



Then the eq. of the line through $A(-\frac{1}{3}, -\frac{2}{3}, 0)$ & $B(0, -\frac{1}{3}, \frac{1}{3})$ is

$$\frac{x + \frac{1}{3}}{0 + \frac{1}{3}} = \frac{y + \frac{2}{3}}{-\frac{1}{3} + \frac{1}{3}} = \frac{z - 0}{\frac{1}{3} - 0}$$

$$\text{or } \frac{x + \frac{1}{3}}{\frac{1}{3}} = \frac{y + \frac{2}{3}}{-\frac{2}{3}} = \frac{z}{\frac{1}{3}}$$

$$\text{or } \frac{x + \frac{1}{3}}{1} = \frac{y + \frac{2}{3}}{-2} = \frac{z}{1}$$

is req. eq. of line in symmetric form.

Q3 Show that the lines $L: x+2y-z-7=0 = y+z-2x-6$ & $M: 3x+6y-3z-8=0 = 2x-y-z$ are parallel.

Sol Given lines are

$$L: x+2y-z-7=0 = -2x+y+z-6$$

$$M: 3x+6y-3z-8=0 = 2x-y-z$$

Let l_1, m_1, n_1 are the d.c.s. of the line L , since it lies on both planes so by condition of perpendicularity

$$\left. \begin{aligned} l_1 + 2m_1 - n_1 &= 0 \\ -2l_1 + m_1 + n_1 &= 0 \end{aligned} \right\}$$

$$\frac{l_1}{2+1} = \frac{-m_1}{1-2} = \frac{n_1}{1+4}$$

$$\frac{l_1}{3} = \frac{m_1}{1} = \frac{n_1}{5}$$

So d.s. of L are $\lambda_1, \mu_1, \nu_1 = 3, 1, 5$

Now

let l_2, m_2, n_2 be the d.s. of line M. Since it lies on both planes so by Condition of perpendicularity

$$\left. \begin{aligned} 3l_2 + 6m_2 - 3n_2 &= 0 \\ 2l_2 - m_2 - n_2 &= 0 \end{aligned} \right\}$$

$$\frac{l_2}{-6-3} = \frac{-m_2}{-3+6} = \frac{n_2}{-3-12}$$

$$\frac{l_2}{-9} = \frac{m_2}{-3} = \frac{n_2}{-15}$$

So d.s. of M are $\lambda_2, \mu_2, \nu_2 = -9, -3, -15$

$$\text{Since } \frac{\lambda_1}{\lambda_2} = \frac{\mu_1}{\mu_2} = \frac{\nu_1}{\nu_2}$$

So the two given lines are parallel.

Q4 Show that the lines

$$L: x+2y-1=0 = 2y-z-1$$

$$M: x-y-1=0 = x-2z-3 \quad \text{are perpendicular.}$$

Sol. Given lines are

$$L: x+2y-1=0 = 2y-z-1$$

$$\& M: x-y-1=0 = x-2z-3$$

let l_1, m_1, n_1 be the d.s. of line L. Since it lies on both planes so by Condition of perpendicularity

$$\left. \begin{aligned} l_1 + 2m_1 + 0n_1 &= 0 \\ 0l_1 + 2m_1 - n_1 &= 0 \end{aligned} \right\}$$

$$\frac{l_1}{-2-0} = \frac{-m_1}{-1-0} = \frac{n_1}{2-0}$$

$$\frac{l_1}{-2} = \frac{m_1}{1} = \frac{n_1}{2}$$

$\Rightarrow$ d.r.s. of L are $\lambda_1, \mu_1, \nu_1 = -2, 1, 2$

Now let l_2, m_2, n_2 be the d.r.s. of line M. Since it lies on both planes so by Condition of perpendicularity,

$$\left. \begin{aligned} l_2 - m_2 + 0n_2 &= 0 \\ l_2 + 0m_2 - 2n_2 &= 0 \end{aligned} \right\}$$

$$\frac{l_2}{2-0} = \frac{-m_2}{-2-0} = \frac{n_2}{0+1}$$

$$\frac{l_2}{2} = \frac{m_2}{2} = \frac{n_2}{1}$$

$\therefore$ d.r.s. of line M are $\lambda_2, \mu_2, \nu_2 = 2, 2, 1$

$$\begin{aligned} \text{Now } \lambda_1 \lambda_2 + \mu_1 \mu_2 + \nu_1 \nu_2 &= (-2)(2) + (1)(2) + (2)(1) \\ &= -4 + 2 + 2 \\ &= 0 \end{aligned}$$

Hence L & M are perp. to each other.

Q5 Find the eq. of the st. line through the pt. $(1, 2, 3)$ & parallel to the line

$$x - y + 2z - 5 = 0 = 3x + y + z + 6$$

Sol. Given eq. of line is

$$x - y + 2z - 5 = 0 = 3x + y + z + 6$$

let l, m, n be the d.c. of this line. Since it lies on both planes so by condition of perpendicularity

$$\left. \begin{aligned} l - m + 2n &= 0 \\ 3l + m + n &= 0 \end{aligned} \right\}$$

$$\frac{l}{-1-2} = \frac{-m}{1-6} = \frac{n}{1+3}$$

$$\frac{l}{-3} = \frac{m}{5} = \frac{n}{4}$$

So d.c. of given line are $\lambda, \mu, \nu = -3, 5, 4$

let L be the req. line.

Since $L \parallel$ given line

So d.c. of req. line L are $-3, 5, 4$

Hence req. eq. of L through $(1, 2, 3)$ & having d.c. $-3, 5, 4$ is

$$\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

Q6 Find the eq. of the planes through the st. line $x + y - z = 0 = 2x - y + 3z - 5$ & perpendicular to the Co-ord. planes.

Sol. Given line is

$$x + y - z = 0 = 2x - y + 3z - 5$$

Then the eq. of plane through given line is

$$(x + y - z) + k(2x - y + 3z - 5) = 0$$

or $(1+2k)x + (1-k)y + (3k-1)z - 5k = 0$ ——— ①

d.s.s. of the normal to plane ① are

$$1+2k, 1-k, 3k-1$$

Case (i)

Since plane ① is perp. to yz -plane whose eq. is

$$x = 0$$

So $(1+2k)(1) = 0$

$$1+2k = 0$$

$$2k = -1$$

$$k = -\frac{1}{2}$$

Put in ①

$$(1+2(-\frac{1}{2}))x + (1+\frac{1}{2})y + (3(-\frac{1}{2})-1)z - 5(-\frac{1}{2}) = 0$$

$$(1-1)x + (\frac{3}{2})y + (-\frac{3}{2}-1)z + \frac{5}{2} = 0$$

$$\frac{3}{2}y - \frac{5}{2}z + \frac{5}{2} = 0$$

$$3y - 5z + 5 = 0 \text{ is req. eq.}$$

Case (ii) Since plane ① is perp. to xz -plane whose

eq. is $y = 0$

So $(1-k) \cdot 1 = 0$

$$k = 1$$

Put in ①

$$(1+2)x + (1-1)y + (3-1)z - 5 = 0$$

$$\text{or } 3x + 2z - 5 = 0$$

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Case (iii)

Since plane ① is $\perp$ to xy -plane whose eq. is

$$z = 0$$

$$\text{So } (3k-1) \cdot 1 = 0$$

$$3k-1 = 0$$

$$3k = 1$$

$$\boxed{k = \frac{1}{3}}$$

Put in ①

$$\left(1+2\left(\frac{1}{3}\right)\right)x + \left(1-\frac{1}{3}\right)y + (1-1)z - 5\left(\frac{1}{3}\right) = 0$$

$$\frac{5}{3}x + \frac{2}{3}y - \frac{5}{3} = 0$$

$$\text{or } \boxed{5x + 2y - 5 = 0} \text{ is req. eq.}$$

Q7 Find eq. of the plane containing the line $x=2t, y=3t, z=4t$ & the intersection of the planes $x+y+z=0$ & $2y-z=0$

Sol: The eq. of a plane through the intersection of planes $x+y+z=0$ & $2y-z=0$ is

$$(x+y+z) + k(2y-z) = 0$$

$$\text{or } x + (1+2k)y + (1-k)z = 0 \quad \text{--- ①}$$

Here d.s. of normal to plane ① are $1, 1+2k, 1-k$

Since this plane contains the line $x=2t, y=3t, z=4t$ whose d.s. are $2, 3, 4$

So by Condition of perpendicularity

$$1(2) + (1+2K)(3) + (1-K)(4) = 0$$

$$2 + 3 + 6K + 4 - 4K = 0$$

$$9 + 2K = 0$$

$$2K = -9$$

$$K = -\frac{9}{2}$$

Put in above eq.

$$(x+y+z) - \frac{9}{2}(2y-z) = 0$$

$$2x + 2y + 2z - 18y + 9z = 0$$

$$2x - 16y + 11z = 0 \text{ is req. eq.}$$

Q8 write an eq. of the family of planes having x-intercept 3, y intercept 2 & a non zero z-intercept. Find the member of the family which is perpendicular to the plane $3x - 2y + z - 4 = 0$

Sol:- Suppose that the non zero z-int. is C

then the eq. of the req. family of planes is

$$\frac{x}{3} + \frac{y}{2} + \frac{z}{C} = 1 \text{ --- (1) where C is parameter}$$

Here d.n.s. of normal to planes are $\frac{1}{3}, \frac{1}{2}, \frac{1}{C}$

If a member of family (1) is perpendicular to plane $3x - 2y + z - 4 = 0$

$$\text{then } 3\left(\frac{1}{3}\right) - 2\left(\frac{1}{2}\right) + 1\left(\frac{1}{C}\right) = 0$$

$$\frac{3}{5} - 1 + \frac{1}{c} = 0$$

$$-\frac{2}{5} + \frac{1}{c} = 0$$

$$\frac{1}{c} = \frac{2}{5}$$

$$\boxed{c = \frac{5}{2}}$$

Put in ①

$$\frac{x}{5} + \frac{y}{2} + \frac{z}{\frac{5}{2}} = 1$$

a $\boxed{\frac{x}{5} + \frac{y}{2} + \frac{2z}{5} = 1}$ is req. member of family

Q9 Find eq. of the plane passing through the pt. $(2, -3, 1)$ & containing the line $x-3 = 2y = 3z-1$

Sol. Given line is

$$x-3 = 2y = 3z-1$$

It can be written as

$$x-2y-3 = 0 = 2y-3z+1$$

Then the eq. of plane containing this line is

$$(x-2y-3) + k(2y-3z+1) = 0$$

Since it passes through $(2, -3, 1)$ so

$$[2-2(-3)-3] + k[2(-3)-3(1)+1] = 0$$

$$(2+6-3) + k(-6-3+1) = 0$$

$$5 + k(-8) = 0$$

$$8K = 5$$

$$K = \frac{5}{8}$$

Put in above eq.

$$(x-2y-3) + \frac{5}{8}(2y-3z+1) = 0$$

$$8x-16y-24+10y-15z+5=0$$

$$8x-6y-15z-19=0 \text{ is req. eq.}$$

Q10 Find eq. of the plane through the line of intersection of the planes $2x-y+3z=0$ & $x+2y-2z-3=0$ and at unit distance from origin.

Sol Eq. of a plane through the line of intersection of given planes is

$$(2x-y+3z) + K(x+2y-2z-3) = 0$$

$$(2+K)x + (2K-1)y + (3-2K)z - 3K = 0$$

Since this plane is at unit distance from 0

$$\text{So } \frac{|(2+K)(0) + (2K-1)(0) + (3-2K)(0) - 3K|}{\sqrt{(2+K)^2 + (2K-1)^2 + (3-2K)^2}} = 1$$

$$\Rightarrow 3K = \sqrt{(2+K)^2 + (2K-1)^2 + (3-2K)^2}$$

Sq. both sides

$$9K^2 = 4 + 4K + K^2 + 4K^2 - 4K + 1 + 9 - 12K + 4K^2$$

$$9/k^2 = 9/k^2 - 12k + 14$$

$$0 = -12k + 14$$

$$12k = 14$$

$$k = \frac{7}{6}$$

Put in above eq.

$$(2x - y + 3z) + \frac{7}{6}(x + 2y - 2z - 3) = 0$$

$$12x - 6y + 18z + 7x + 14y - 14z - 21 = 0$$

$$19x + 8y + 4z - 21 = 0 \text{ is req. eq.}$$

Q11 Find eq. of the perpendicular from the origin to the line $x + 2y + 3z + 4 = 0$ or $2x + 3y + 4z + 5$

Also find the Co-ords. of the foot of perpendicular.

Soln: Given eq. of line is

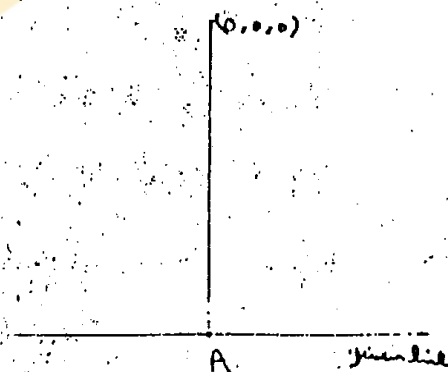
$$\left. \begin{aligned} x + 2y + 3z + 4 &= 0 \\ 2x + 3y + 4z + 5 &= 0 \end{aligned} \right\}$$

First we find its symmetric form.

Let l, m, n be the d.c.s. of given line.

Since it lies on both planes.

So by Condition of perpendicularity



$$\Rightarrow \left. \begin{aligned} l+2m+3n &= 0 \\ 2l+3m+4n &= 0 \end{aligned} \right\}$$

$$\frac{l}{2-1} = \frac{-m}{4-6} = \frac{n}{3-4}$$

$$\frac{l}{-1} = \frac{m}{2} = \frac{n}{-1}$$

$$\text{or } \frac{l}{1} = \frac{m}{-2} = \frac{n}{1}$$

So dir.s. of given line are $1, -2, 1$

To find a pt. on line Put $z = 0$

$$\left. \begin{aligned} x+2y+4 &= 0 \\ 2x+3y+5 &= 0 \end{aligned} \right\}$$

$$\frac{x}{10-12} = \frac{-2}{5-8} = \frac{1}{3-4}$$

$$\frac{x}{-2} = \frac{2}{3} = \frac{1}{-1}$$

$$\Rightarrow x = 2, y = -3$$

So $(2, -3, 0)$ is a pt. on the given line.

Hence the symmetric form of given line is

$$\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z}{1} = t(\text{say}) \quad \text{--- ①}$$

$$\Rightarrow \left. \begin{aligned} x &= 2+t \\ y &= -3-2t \\ z &= t \end{aligned} \right\}$$

Any pt. on this line is $(2+t, -3-2t, t)$

Let A be the foot of perpendicular from origin to given line. Then Co-ords. of pt. A are

$$A(2+t, -3-2t, t)$$

Since line OA is perp. to line ①

$$\text{So } 1(2+t) - 2(-3-2t) + 1(t) = 0$$

$$2+t+6+4t+t=0$$

$$6t+8=0$$

$$3t+4=0$$

$$t = -\frac{4}{3}$$

$$\begin{aligned} \text{So Co-ords. of foot of } \perp \text{ are } A\left(2-\frac{4}{3}, -3+\frac{8}{3}, -\frac{4}{3}\right) \\ = A\left(\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}\right) \end{aligned}$$

Now the req. perp. distance = $|OA|$

$$= \sqrt{\left(\frac{2}{3}-0\right)^2 + \left(-\frac{1}{3}-0\right)^2 + \left(-\frac{4}{3}-0\right)^2}$$

$$= \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{16}{9}}$$

$$= \sqrt{\frac{4+1+16}{9}}$$

$$= \sqrt{\frac{21}{9}}$$

$$= \sqrt{\frac{7}{3}} \quad \text{Ans.}$$

Q12 A variable plane is at a const. distance p from origin & meets the axes in A, B, C .

Through A, B & C , planes are drawn parallel to the Co-ord. planes. Show that the locus of their pt. of intersection is given by $\bar{x}^2 + \bar{y}^2 + \bar{z}^2 = p^2$.

Sol.

Let the eq. of plane is

$$lx + my + nz = p \quad \text{--- (1)}$$

where l, m, n are the d.c.s. of normal to plane (1)

Now eq. (1) can be written as

$$\frac{lx}{p} + \frac{my}{p} + \frac{nz}{p} = 1$$

$$\text{or } \frac{\frac{x}{\frac{p}{l}}}{1} + \frac{\frac{y}{\frac{p}{m}}}{1} + \frac{\frac{z}{\frac{p}{n}}}{1} = 1$$

which is intercept form of plane (1)

Hence Co-ords. of pts. A, B & C are

$$A\left(\frac{p}{l}, 0, 0\right), B\left(0, \frac{p}{m}, 0\right) \text{ \& } C\left(0, 0, \frac{p}{n}\right)$$

Now eq. of a plane through pt. $A\left(\frac{p}{l}, 0, 0\right)$ & $\parallel$ to yz -plane is $x = \frac{p}{l}$

eq. of plane through pt. $B\left(0, \frac{p}{m}, 0\right)$ & $\parallel$ to xz -plane is $y = \frac{p}{m}$

Similarly, eq. of plane through $C\left(0, 0, \frac{p}{n}\right)$ & $\parallel$ to xy -plane is $z = \frac{p}{n}$

clearly the pt. of intersection of given planes is 105

$$(x, y, z) = \left(\frac{p}{l}, \frac{p}{m}, \frac{p}{n}\right)$$

$$\Rightarrow x = \frac{p}{l}, y = \frac{p}{m}, z = \frac{p}{n}$$

$$\Rightarrow \left. \begin{aligned} l &= \frac{p}{x} \\ m &= \frac{p}{y} \\ n &= \frac{p}{z} \end{aligned} \right\}$$

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Since $l^2 + m^2 + n^2 = 1$

$$\frac{p^2}{x^2} + \frac{p^2}{y^2} + \frac{p^2}{z^2} = 1$$

$$\text{or } \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$$

$$\text{or } \boxed{\frac{x^{-2}}{1} + \frac{y^{-2}}{1} + \frac{z^{-2}}{1} = p^{-2}} \text{ is req. locus.}$$

Q13 let A, B, C be the pts. as in problem 12,
Prove that the locus of the Centroid of the
tetrahedron OABC is

$$\vec{x}^2 + \vec{y}^2 + \vec{z}^2 = 16\vec{p}^2, \text{ O being origin.}$$

Sol. Since the Co-ords. of pts. A, B & C are

$$A\left(\frac{p}{l}, 0, 0\right), B\left(0, \frac{p}{m}, 0\right) \text{ \& } C\left(0, 0, \frac{p}{n}\right).$$

Now Co-ords. of pt. O are O(0, 0, 0)

Hence the Co-ords. of the vertices of tetrahedron^{1.6}
OABC are $O(0,0,0)$, $A(\frac{p}{4x}, 0, 0)$, $B(0, \frac{p}{4y}, 0)$, $C(0, 0, \frac{p}{4z})$

Now Centroid of Tetrahedron is $(\frac{p}{4x}, \frac{p}{4y}, \frac{p}{4z})$

$$\Rightarrow \left. \begin{aligned} x &= \frac{p}{4x} \\ y &= \frac{p}{4y} \\ z &= \frac{p}{4z} \end{aligned} \right\}$$

$$\text{or } \left. \begin{aligned} l &= \frac{p}{4x} \\ m &= \frac{p}{4y} \\ n &= \frac{p}{4z} \end{aligned} \right\}$$

$$\therefore l^2 + m^2 + n^2 = 1$$

$$\frac{p^2}{16x^2} + \frac{p^2}{16y^2} + \frac{p^2}{16z^2} = 1$$

$$\frac{p^2}{x^2} + \frac{p^2}{y^2} + \frac{p^2}{z^2} = 16$$

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$$

$$\text{or } \boxed{\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 16p^{-2}}$$

which is req. locus.